

CampusEats — Design

Tasks 1, 3, 4 and 6 — Capabilities, Service Contracts, Central Operation & Service Validation

Team / Academic Details	Information
Course	CS 543 Web Services
Project	CampusEats
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Included Tasks	1, 3, 4 and 6

This document is prepared from the CampusEats capability list, service contracts, placeOrder specification, and the supplied CampusEats service design diagram.

Task 1 — Capability List

The following capabilities are included for the CampusEats system.

No.	Capability	What the system does
1	Register / Login	Students can create an account and log in to the system.
2	Search Food Vendors	Students can search for food shops available on campus.
3	View Menu	Students can see available food items and their prices.
4	Add Food to Cart	Students can add the food they want to order to the cart.
5	Place Order	Students can place their food order.
6	Make Payment	Students can pay for their food order.
7	Track Order	Students can check the status of their order.
8	Cancel Order	Students can cancel an order when cancellation is allowed.
9	Manage Menu	Vendors can add, update, or remove food items.
10	Manage Orders	Vendors can receive and process student orders.
11	Give Rating and Review	Students can give ratings and reviews for their orders.
12	Manage Users	The administrator can manage users and their access.

Task 3 — Service Contracts

Each service exposes operations that other services may call.

3.1 User & Auth Service

Operation	Input	Output	Errors
verifyUser	User login information	Verification result	Invalid credentials; User not found
checkRole	User information	User role	User not found; Unauthorized

3.2 Catalogue Service

Operation	Input	Output	Errors
checkItem	Food item information	Item availability	Item not found; Item unavailable
updateAvailability	Food item information and availability status	Updated availability confirmation	Item not found
updateMenu	Menu information	Menu update confirmation	Invalid menu information; Update failed

3.3 Cart & Order Service

Operation	Input	Output	Errors
placeOrder	User, cart and payment information	Order confirmation and status	Empty cart; Item unavailable; Payment failed
trackOrder	Order information	Current order status	Order not found
cancelOrder	Order information	Cancellation confirmation	Order not found; Order cannot be cancelled
checkOrder	Order information	Order eligibility/status	Order not found

3.4 Payment Service

Operation	Input	Output	Errors
makePayment	Order and payment information	Payment confirmation and status	Payment failed; Payment declined

3.5 Review Service

Operation	Input	Output	Errors
submitReview	Order information, rating and comment	Review confirmation	Invalid rating; Review not allowed

3.6 Admin & Vendor Service

Operation	Input	Output	Errors
manageVendor	Vendor information	Vendor management confirmation	Vendor not found; Invalid information

Task 4 — Central Operation: placeOrder

Purpose: To place a food order for a user using the selected cart items and complete the required payment.

4.1 Inputs

Input	Description
User information	Information needed to identify and verify the user placing the order.
Cart information	The selected food items and their quantities.
Payment information	The payment method and information needed to complete payment.

4.2 Outputs

Output	Description
Order confirmation	Confirmation that the order was successfully placed.
Order status	The current status of the new order.
Payment status	The result of the payment operation.

4.3 Error Cases

Error Case	Meaning
User not verified	The user cannot be verified.
Cart is empty	There are no items to order.
Item unavailable	One or more selected items are unavailable.
Payment failed	The payment could not be completed.
Order cannot be placed	The order could not be completed because of a processing or service failure.

4.4 Internal Details Hidden from Callers

Hidden detail	How it works internally
Order storage	The service keeps order information internally and manages order status without exposing how the information is stored.
Cart processing	The service reads selected items and quantities from the cart and checks that the cart can be converted into an order.
Amount calculation	The service calculates the amount for the selected order, including applicable charges defined by the system.
Item availability	The service checks with Catalogue Service that selected items are available before confirming the order.
Payment processing	The service sends the required payment request to Payment Service and uses the returned payment result to decide whether the order can continue.
Order status update	After successful processing, the service changes the order to the appropriate status and returns that status.
Communication with other services	The caller does not need to know how Cart & Order communicates with User & Auth, Catalogue, or Payment.
Database implementation	Database tables, SQL queries, primary keys, foreign keys and ID formats are hidden implementation details.

4.5 Functionality of placeOrder

Step 1 — Verify user: Make sure the user is valid and allowed to place an order.

Step 2 — Validate cart: Check that the cart contains items and that selected quantities are valid.

Step 3 — Check availability: Ask Catalogue Service whether the selected items are currently available.

Step 4 — Calculate amount: Determine the amount that needs to be paid for the selected order.

Step 5 — Process payment: Request payment through Payment Service; if payment fails, the order is not confirmed.

Step 6 — Confirm order: After successful payment and validation, confirm the order and update the required status.

Step 7 — Return result: Return order confirmation, order status and payment status to the caller.

Task 6 — Service Validation

Each service is checked against the five required service properties: reachable, self-contained, has a contract, independent and loosely coupled. The validation follows the supplied CampusEats service design diagram.

Service	Reachable	Self-contained	Contract	Independent	Loosely coupled
User & Auth	Yes — Cart & Order calls verifyUser; Admin & Vendor calls checkRole.	Yes — owns User data.	Yes — verifyUser, checkRole.	Yes — no shared data is shown.	Yes — communication is through operations.
Catalogue	Yes — Cart & Order calls checkItem/updateAvailability; Admin & Vendor calls updateMenu.	Yes — owns Menu data.	Yes — checkItem, updateAvailability, updateMenu.	Yes — owns its data and has its own interface.	Yes — callers use operations rather than its data.
Cart & Order	Yes — central order service and exposes order operations.	Yes — owns Cart and Order data.	Yes — placeOrder, trackOrder, cancelOrder, checkOrder.	Yes — dependencies exist, but no shared database is shown.	Yes — dependencies are expressed through contracts.
Payment	No — the supplied diagram does not show a valid incoming service call to Payment.	Yes — owns Payment data.	Yes — makePayment.	Yes — its data is separately owned.	Yes — the service boundary is separate and no shared data is shown.
Review	Yes — the supplied diagram has an incoming connection to Review.	Yes — owns Review data.	Yes — submitReview.	Yes — owns its data.	Yes — interaction is through service operations.
Admin & Vendor	Yes — service exists for admin/vendor functions.	Yes — owns Vendor and Admin data.	Yes — manageVendor.	Yes — no shared data is shown.	Yes — cross-service communication uses operations.

Validation summary: The services are generally self-contained, contract-based, independent in data ownership, and loosely coupled. The supplied diagram shows a reachability issue for the Payment Service, while the other service boundaries and contracts are represented separately.

Consistency with placeOrder: The placeOrder operation requires user verification, item availability checking and payment processing through the appropriate service contracts.